SALES ANALYTICS & BUSINESS PERFORMANCE REPORT

Adventure Works Business Intelligence Solution

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Tools Used: Power BI, DAX, Power Query, Data Modeling

Executive Summary

Adventure Works is a retail and manufacturing business that sells a wide range of products across multiple product categories, customer segments, and geographic locations.

As the business grows, management requires timely access to accurate information to understand sales performance, customer behavior, product profitability, and growth opportunities. To address this need, I developed an interactive Power BI Sales Analytics Dashboard that consolidates sales data into a single decision-support platform.

The solution enables stakeholders to monitor key business metrics, analyze customer purchasing patterns, evaluate product performance, identify regional trends, and investigate factors influencing sales outcomes.

The dashboard was designed to answer critical business questions such as:

- How much revenue is the business generating?
- Which products drive the highest sales? (Orders & Returns)
- Which customers contribute the most value?
- Which regions perform best?
- What factors influence business performance?
- Where should management focus future investments?

The project demonstrates practical application of Business Intelligence principles, data modeling, KPI development, interactive reporting, and business analytics using Power BI.

Business Problem

Adventure Works generates thousands of sales transactions across different product categories, customer groups, and geographical markets. Without a centralized reporting solution, decision-makers may struggle to:

- Monitor revenue performance and identify high-performing products
- Understand customer purchasing behavior and evaluate regional performance
- Detect business opportunities and risks

The objective of this project was therefore to transform raw transactional data into meaningful business insights that support strategic and operational decision-making.

Project Objectives

1. **Monitor Business Performance:** Provide management with a high-level view of overall performance.
2. **Track Key Performance Indicators:** Develop KPI metrics that can be monitored regularly.
3. **Analyze Customer Behavior:** Identify customer segments and purchasing trends.
4. **Evaluate Product Performance:** Determine which products and categories contribute most to revenue.
5. **Assess Regional Performance:** Understand how sales vary across locations.
6. **Support Data-Driven Decisions:** Provide actionable insights for business growth.

Data Preparation & Modeling

Before analysis, the data was prepared using Power Query. A star schema was implemented to improve performance and enable efficient analytical reporting.

Activities Performed:

- Data cleaning and Validation
- Relationship creation
- Data type correction
- Model optimization

Star Schema Structure:

- **Fact Tables:** Sales Data, Return Data
- **Dimension Tables:** Customer, Product, Categories, Subcategories, Calendar, Territory

Key Performance Indicators (KPIs)

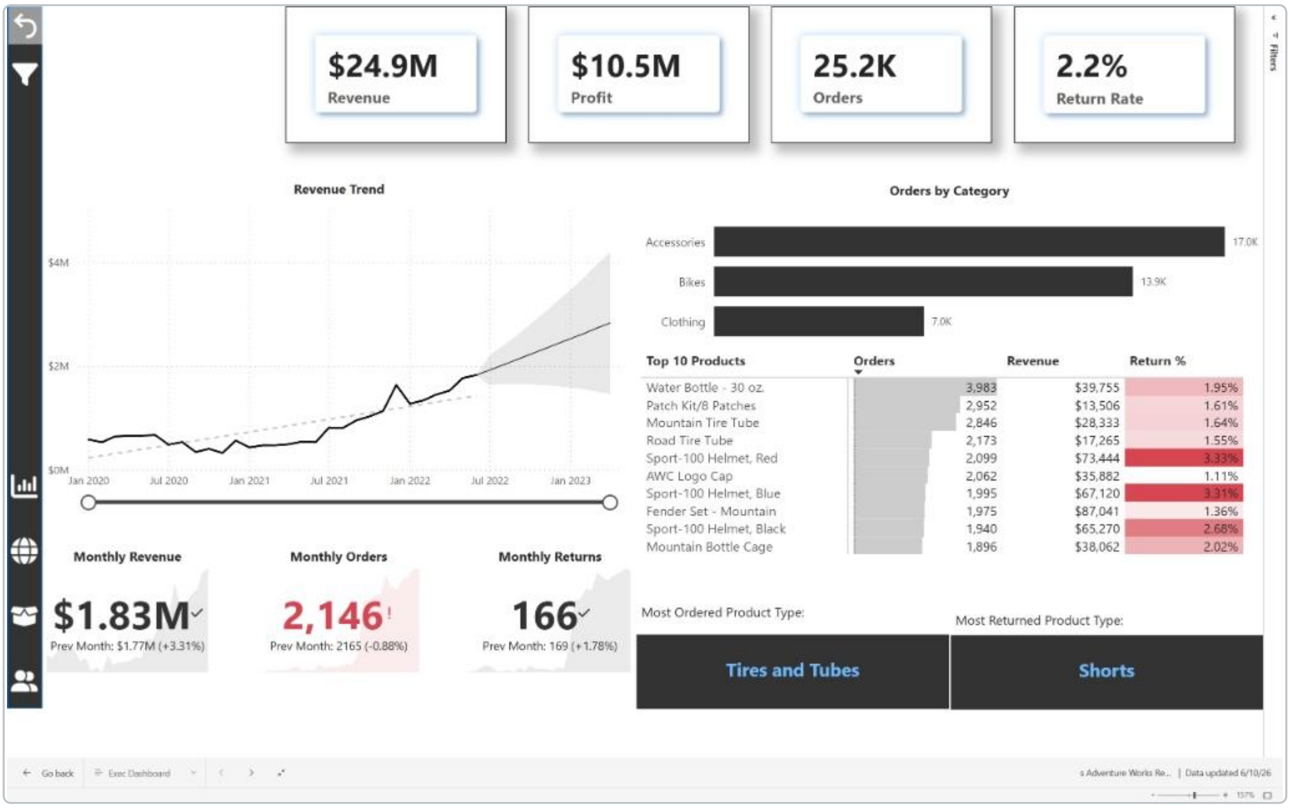
The dashboard focuses on the following business metrics. These KPIs help management quickly assess business health and performance.

KPI	Purpose
Total Revenue	Measures overall sales performance
Total Orders	Measures transaction volume
Total Returns	Tracks product returns
Total Customers	Measures customer reach
Revenue Growth	Tracks business growth trends
Return Rate	Measures product return performance

Dashboard Walkthrough & Key Findings

1. Executive Performance

The business demonstrates strong top-line health, generating **$24.9M** in total revenue and **$10.5M** in profit across **25.2K** orders. We maintained a sustainable overall return rate of **2.2%**. Recent monthly trends indicate steady growth, with the most recent month closing at **$1.83M** (a 3.31% increase from the previous period).

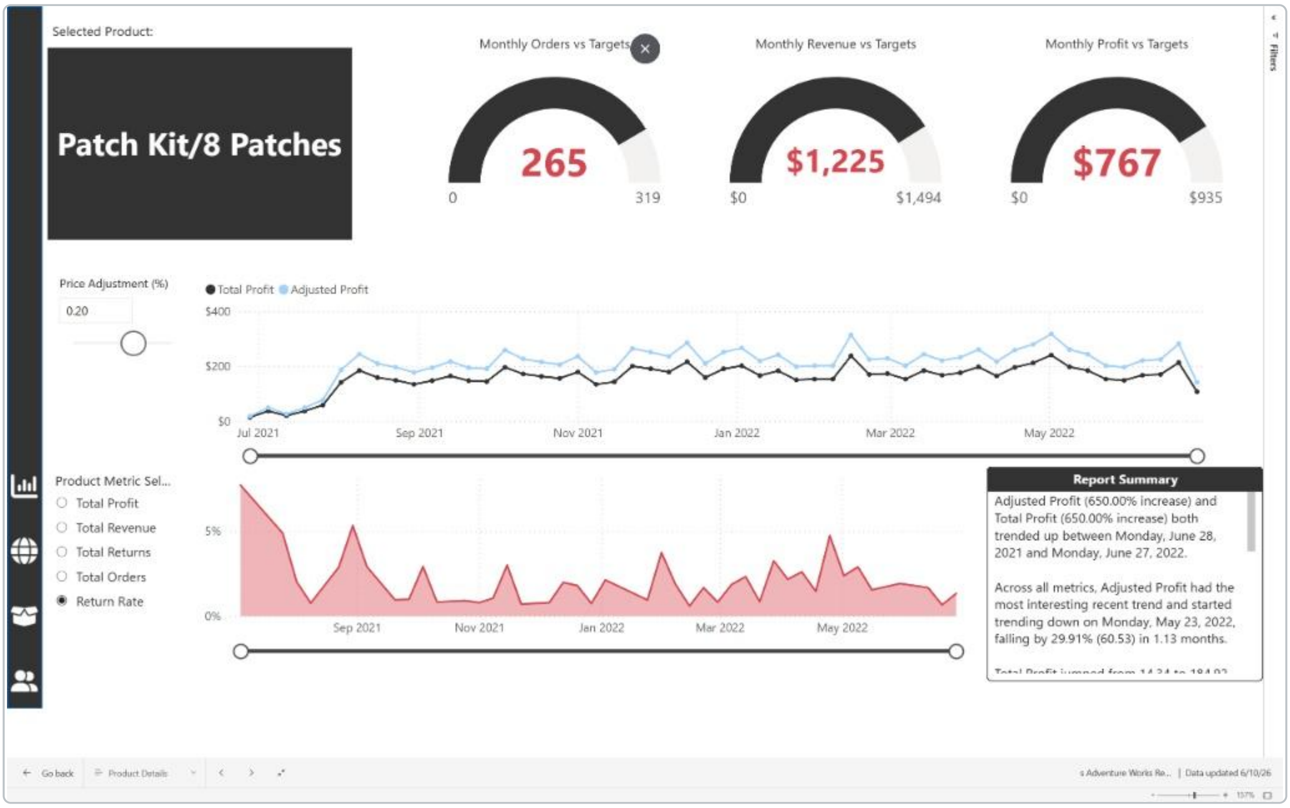


2. Product & Category Dynamics

Performance is highly concentrated. While the Accessories category drives the highest transaction volume (**17.0K** orders), Bikes remain the core revenue driver (**13.9K** orders).

- **Revenue Drivers:** Specific high-ticket items disproportionately impact the top line. For example, the Mountain Fender Set generated **$87,041** in revenue.
- **Volume Drivers:** The 30 oz. Water Bottle acts as a primary entry-point product, driving nearly **4,000** individual orders.

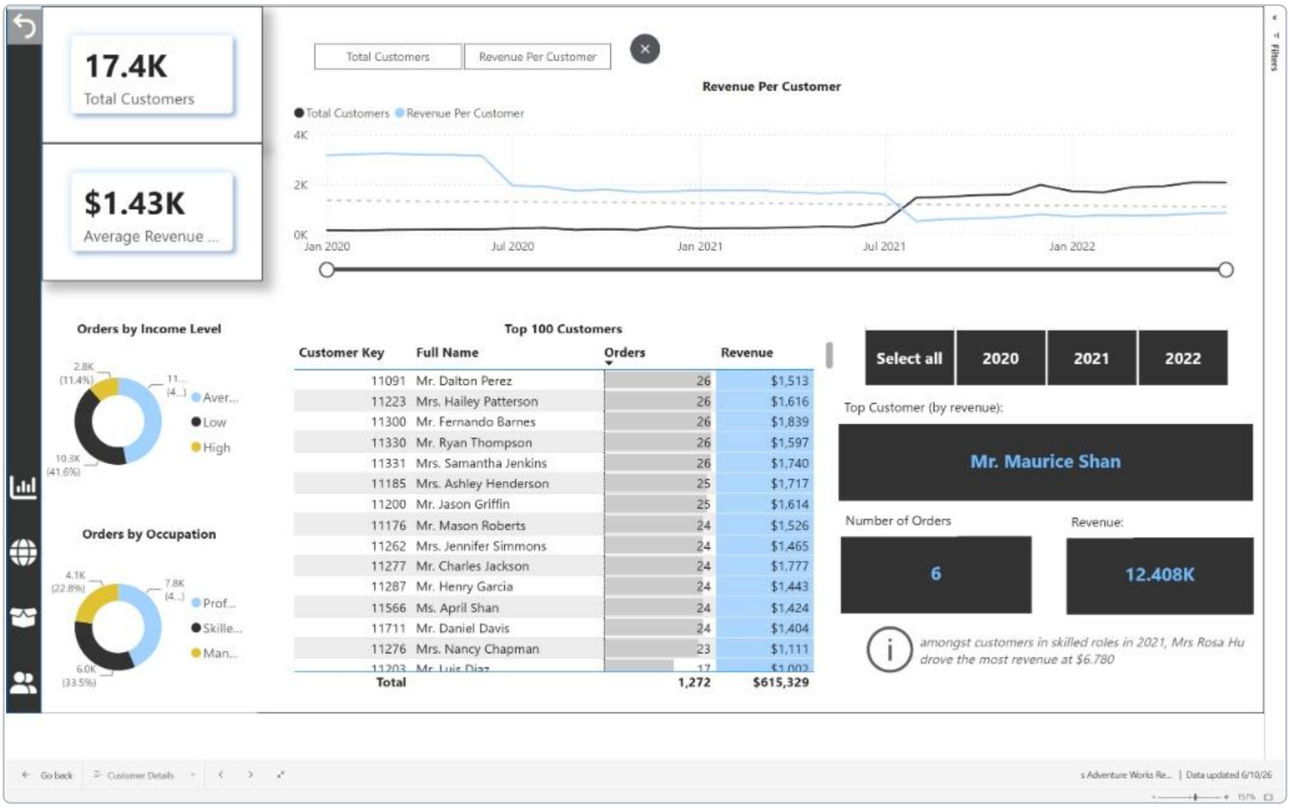

Back to report
ORDERS BY CATEGORY
Filters
Accessories
17.0K
Bikes
13.9K
Clothing
7.0K
Go back
Exec Dashboard
s Adventure Works Re...
Data updated 6/10/26
100%

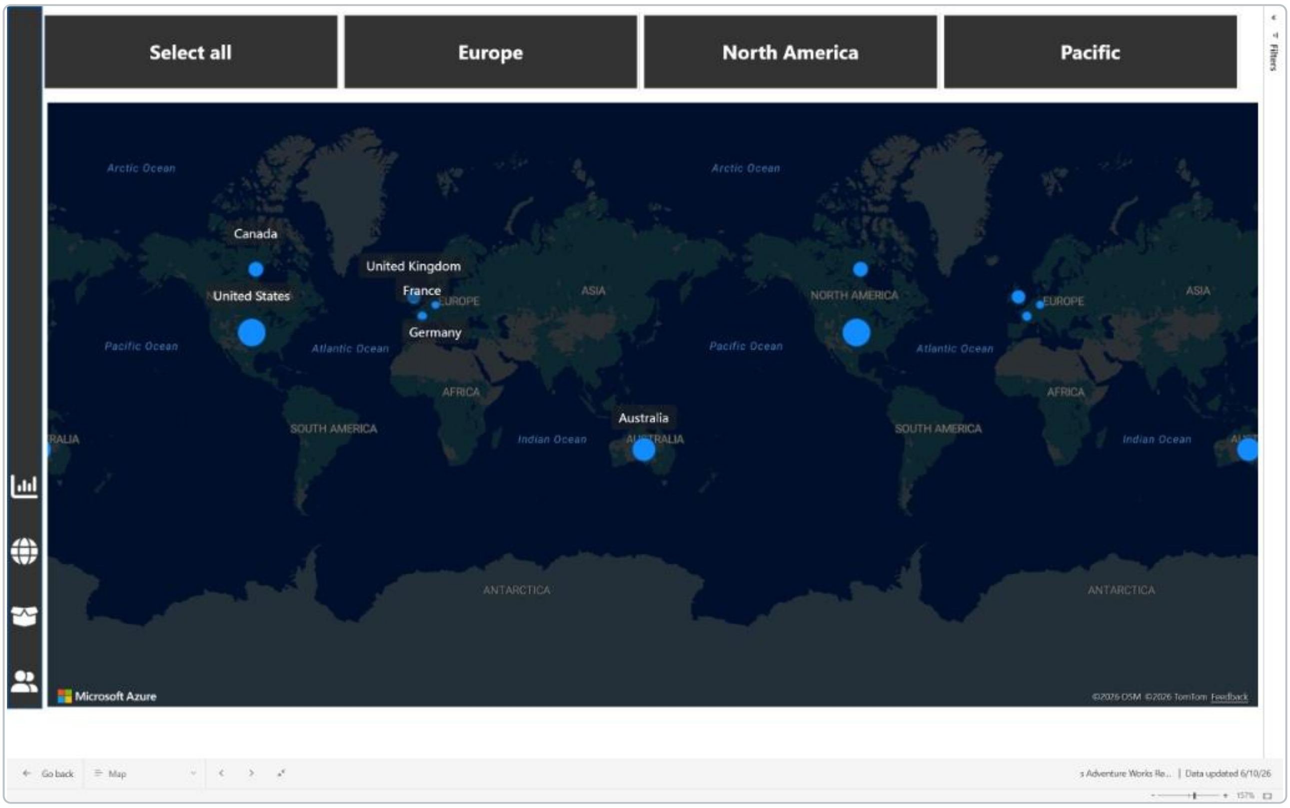

Selected Product:
Patch Kit/8 Patches
Monthly Orders vs Targets
265
0
319
Monthly Revenue vs Targets
$1,225
$0
$1,494
Monthly Profit vs Targets
$767
$0
$935
Filters
Price Adjustment (%)
0.20
Total Profit
Adjusted Profit
$400
$200
$0
Jul 2021
Sep 2021
Nov 2021
Jan 2022
Mar 2022
May 2022
Product Metric Sel...
Total Profit
Total Revenue
Total Returns
Total Orders
Return Rate
5%
0%
Sep 2021
Nov 2021
Jan 2022
Mar 2022
May 2022
Report Summary
Adjusted Profit (650.00% increase) and Total Profit (650.00% increase) both trended up between Monday, June 28, 2021 and Monday, June 27, 2022.
Across all metrics, Adjusted Profit had the most interesting recent trend and started trending down on Monday, May 23, 2022, falling by 29.91% (60.53) in 1.13 months.
Go back
Product Details
s Adventure Works Re...
Data updated 6/10/26

3. Customer Value, Segmentation & Geography

Our active customer base consists of **17.4K** individuals, yielding an average revenue per customer of **$1.43K**. However, customer value is not evenly distributed. A segment of high-tier clients vastly outperforms the average—for instance, our top customer generated over **$12.4K** from just 6 orders.

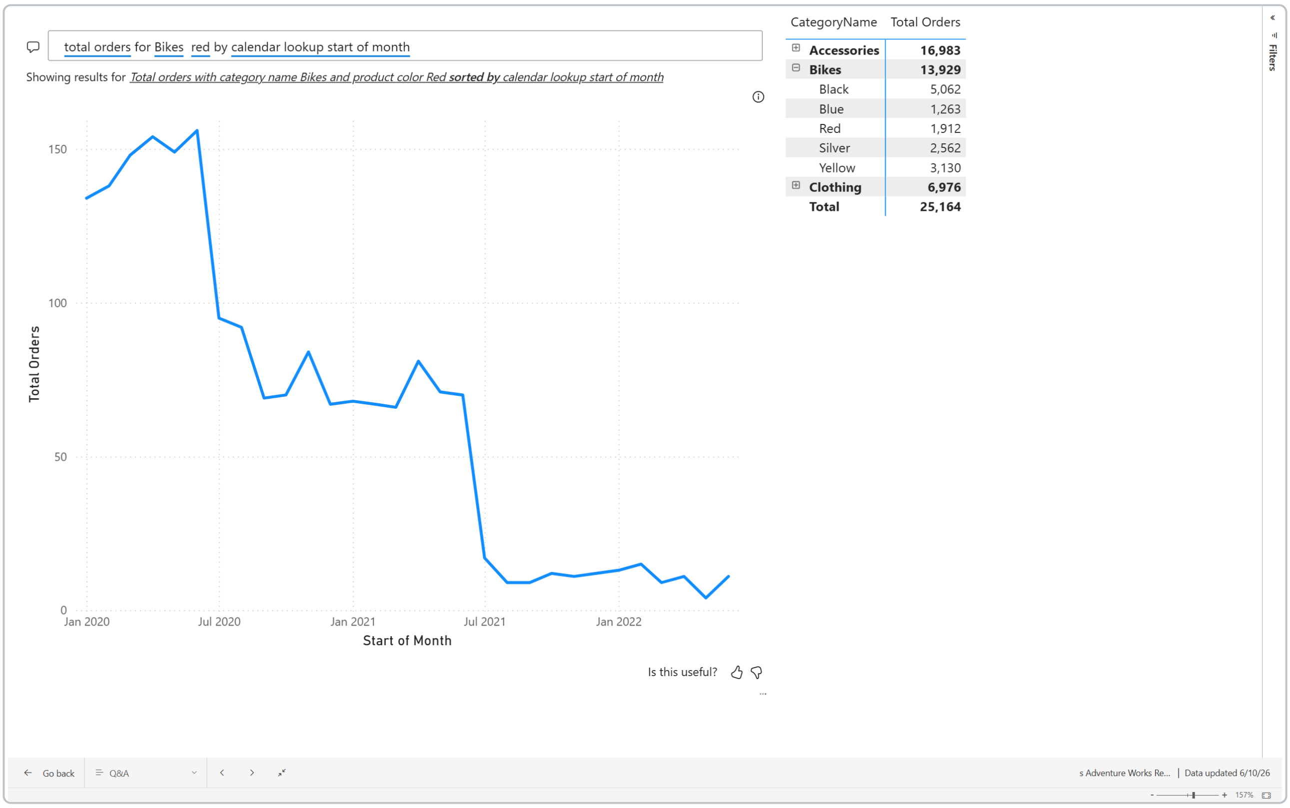
Furthermore, mapping these sales geographically reveals distinct regional performance differences, allowing us to pinpoint where our high-value segments are clustered.

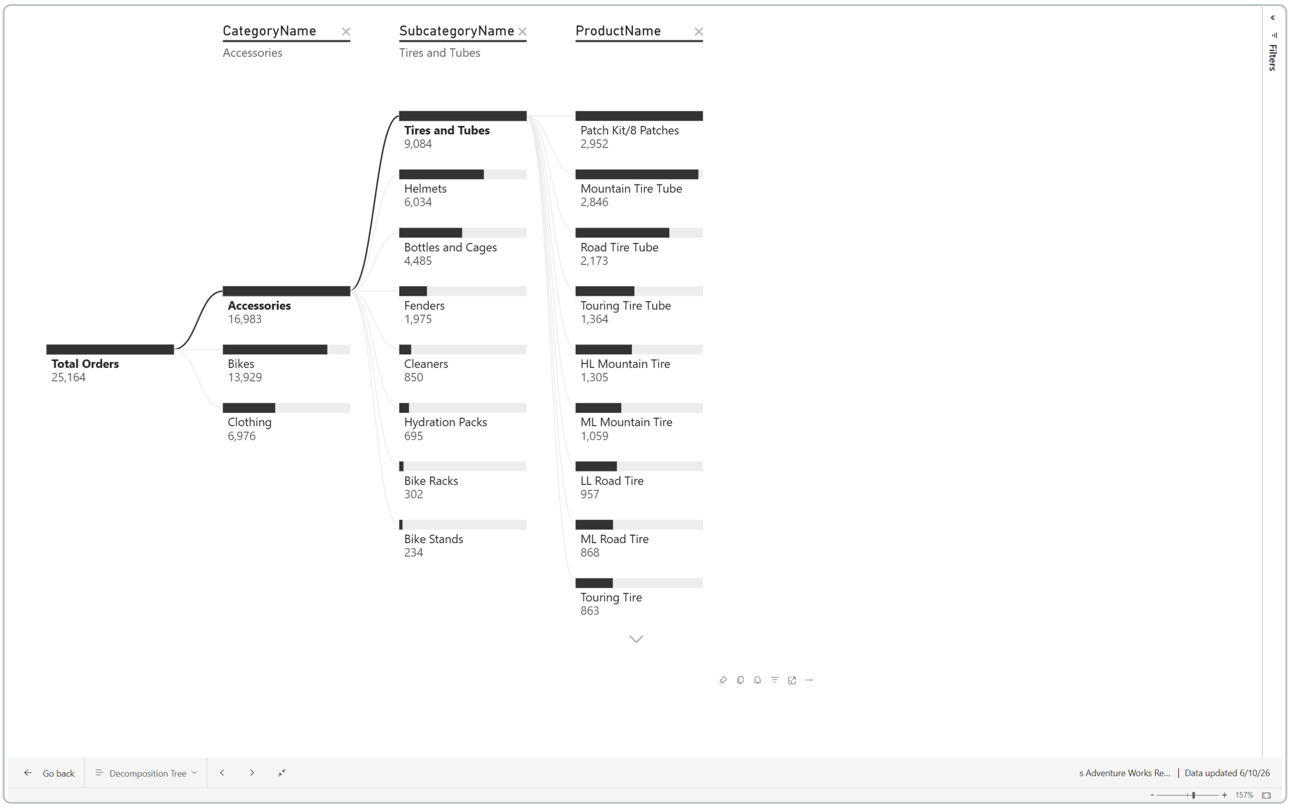

17.4K
Total Customers
$1.43K
Average Revenue ...
Total Customers
Revenue Per Customer
Revenue Per Customer
Total Customers Revenue Per Customer
4K
2K
0K
Jan 2020
Jul 2020
Jan 2021
Jul 2021
Jan 2022
Orders by Income Level
Aver...
Low
High
Orders by Occupation
Prof...
Skille...
Man...
Top 100 Customers
Customer Key Full Name Orders Revenue
11091 Mr. Dalton Perez 26 $1,513
11223 Mrs. Hailey Patterson 26 $1,616
11300 Mr. Fernando Barnes 26 $1,839
11330 Mr. Ryan Thompson 26 $1,597
11331 Mrs. Samantha Jenkins 26 $1,740
11185 Mrs. Ashley Henderson 25 $1,717
11200 Mr. Jason Griffin 25 $1,614
11176 Mr. Mason Roberts 24 $1,526
11262 Mrs. Jennifer Simmons 24 $1,465
11277 Mr. Charles Jackson 24 $1,777
11287 Mr. Henry Garcia 24 $1,443
11566 Ms. April Shan 24 $1,424
11711 Mr. Daniel Davis 24 $1,404
11276 Mrs. Nancy Chapman 23 $1,111
Total 1,272 $615,329
Select all
2020
2021
2022
Top Customer (by revenue):
Mr. Maurice Shan
Number of Orders
Revenue:
6
12.408K
amongst customers in skilled roles in 2021, Mrs Rosa Hu drove the most revenue at $6.780
Go back
Customer Details
Data updated 6/10/26
Filters


Select all
Europe
North America
Pacific
Arctic Ocean
Canada
United States
United Kingdom
France
Germany
EUROPE
ASIA
Pacific Ocean
Atlantic Ocean
AFRICA
SOUTH AMERICA
Indian Ocean
Australia
NORTH AMERICA
ANTARCTICA
Microsoft Azure
Go back
Map
Data updated 6/10/26
Filters

4. Root Cause & Self-Service Analytics

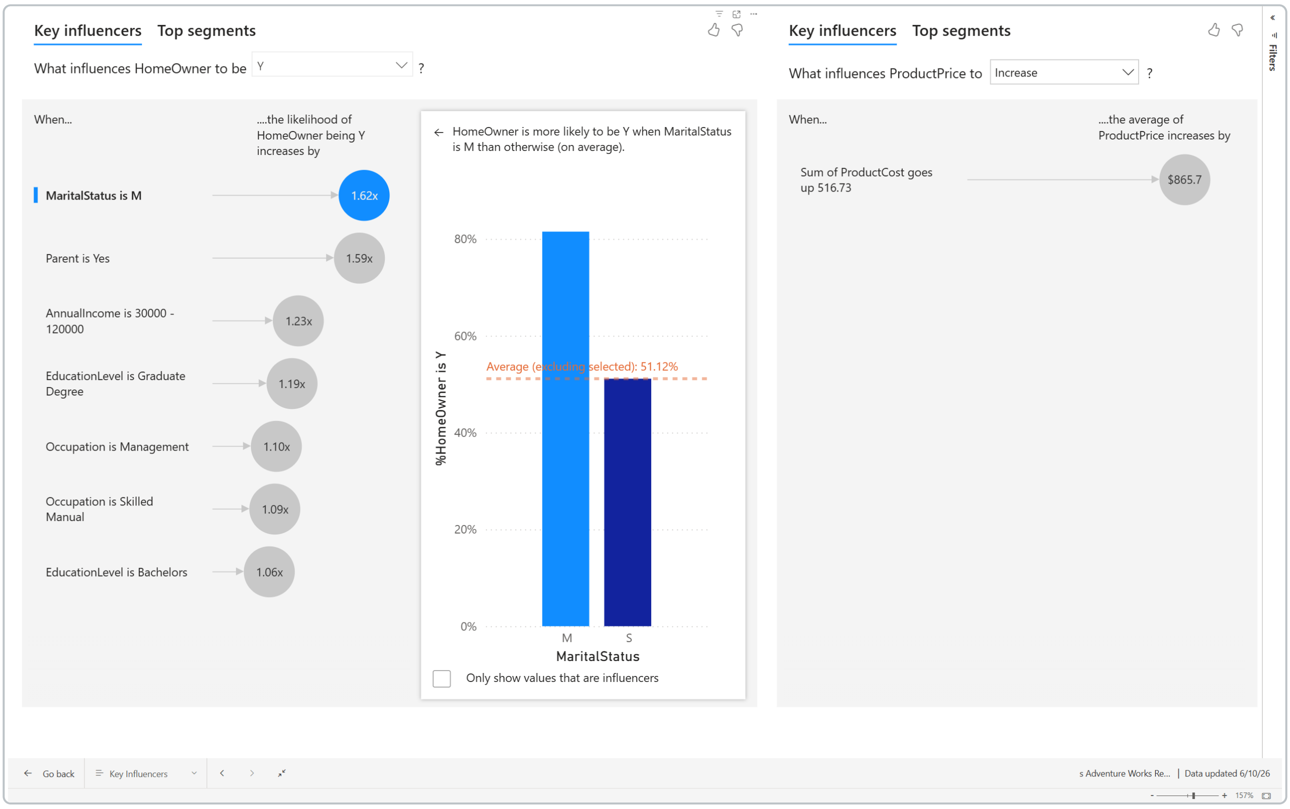
To allow stakeholders to ask questions organically (e.g., "Total orders for bikes red by start of month"), I integrated natural language Q&A features. Additionally, the Decomposition Tree allows us to break down exactly *why* a metric is performing the way it is by drilling directly from Category down to the specific Customer.


total orders for Bikes red by calendar lookup start of month
Showing results for Total orders with category name Bikes and product color Red sorted by calendar lookup start of month
Total Orders
150
100
50
0
Jan 2020
Jul 2020
Jan 2021
Jul 2021
Jan 2022
Start of Month
Is this useful?
CategoryName
Total Orders
Accessories 16,983
Bikes 13,929
Black 5,062
Blue 1,263
Red 1,912
Silver 2,562
Yellow 3,130
Clothing 6,976
Total 25,164
Go back
Q&A
Filters
s Adventure Works Re... | Data updated 6/10/26


CategoryName
Accessories
SubcategoryName
Tires and Tubes
ProductName
Total Orders
25,164
Accessories
16,983
Bikes
13,929
Clothing
6,976
Tires and Tubes
9,084
Helmets
6,034
Bottles and Cages
4,485
Fenders
1,975
Cleaners
850
Hydration Packs
695
Bike Racks
302
Bike Stands
234
Patch Kit/8 Patches
2,952
Mountain Tire Tube
2,846
Road Tire Tube
2,173
Touring Tire Tube
1,364
HL Mountain Tire
1,305
ML Mountain Tire
1,059
LL Road Tire
957
ML Road Tire
868
Touring Tire
863
Go back
Decomposition Tree
Filters
s Adventure Works Re... | Data updated 6/10/26

5. Key Influencers & Demographics

Using the Key Influencers analysis, we identified that demographic factors significantly impact purchasing power. For example, customers with a Marital Status of "M" (Married) are **1.62x** more likely to be homeowners, which historically correlates with the stability and disposable income required for high-ticket purchases like our top-tier bikes.



Strategic Recommendations

Based on the data models and analysis, I recommend the following operational adjustments:

- **Implement High-Tier Retention Protocols:** With average customer revenue at $1.43K, losing a top-tier customer (spending $10K+) has a disproportionate impact. We should deploy VIP retention strategies and dedicated account management for the top 5% of our customer base.
- **Investigate Clothing Quality Control:** Initiate a supplier audit for the *Shorts* product line to identify the root cause of the high return rate, which is actively eroding profit margins in the Clothing category.
- **Optimize Inventory for Volume vs. Revenue:** Treat volume drivers (like Water Bottles) as acquisition tools to ensure they are never out of stock, while focusing marketing spend on high-margin revenue drivers (like Fender Sets and Bikes).

- **Targeted Geographic Campaigns:** Leverage the demographic data and geographic clustering to refine our marketing spend, targeting regions with high concentrations of our most profitable buyer personas.

Skills Demonstrated Through This Project

- Business Intelligence & Analytics
- KPI Design & Dashboard Development
- Power BI, DAX Measures, Power Query
- Data Modeling & Data Storytelling
- Interactive Visualizations
- Customer & Product Analytics
- Geographic Analysis
- Root Cause Analysis
- Insight Generation & Reporting
- Strategic Recommendations

Conclusion

The Adventure Works Sales Analytics Dashboard successfully transforms transactional sales data into actionable business intelligence. The solution provides management with visibility into sales performance, customer behavior, product trends, and business drivers while supporting informed decision-making through interactive analytics.

Beyond dashboard development, this project demonstrates the ability to translate raw data into meaningful insights and practical recommendations that can influence business outcomes.